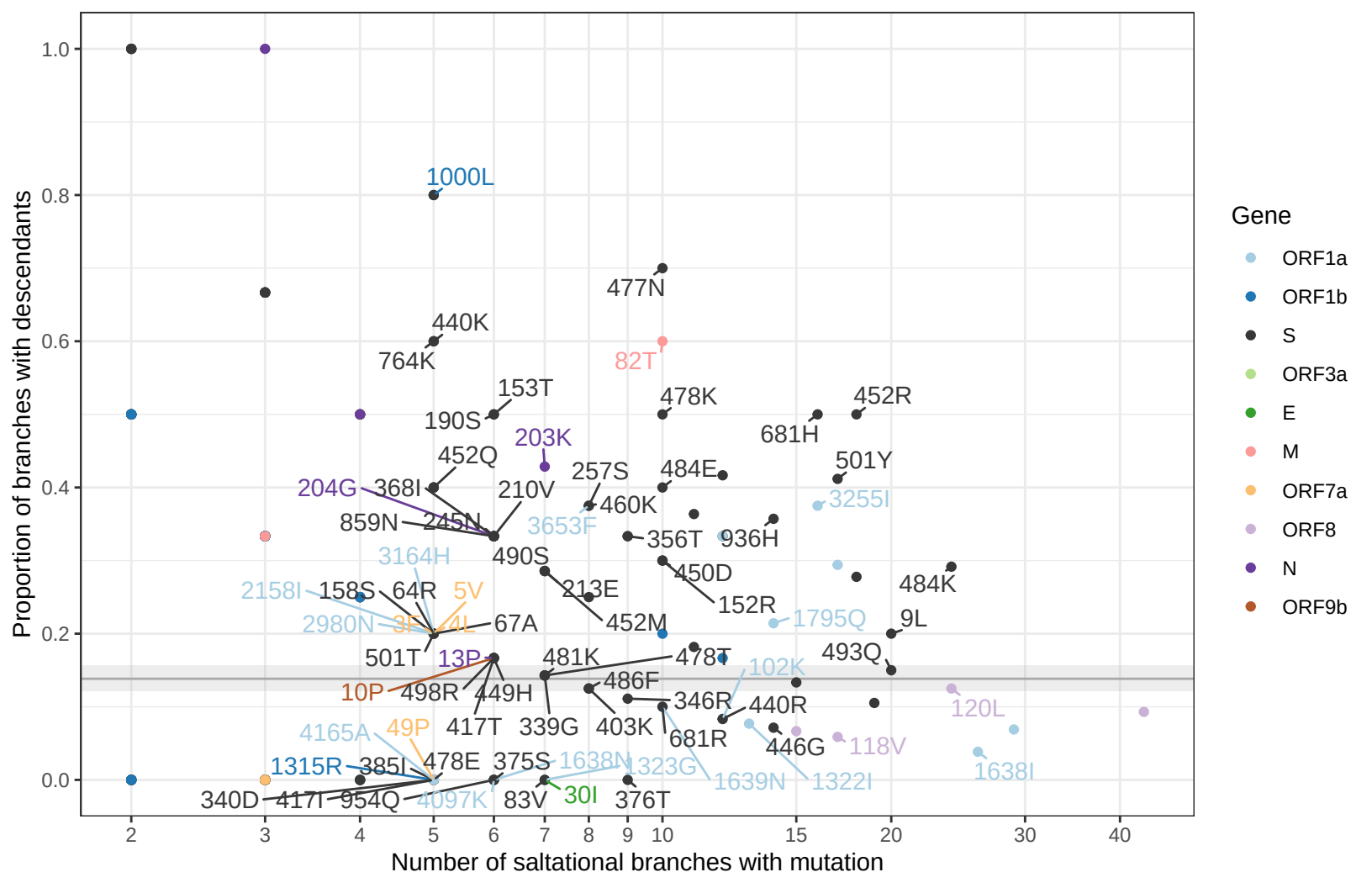
Proportion of branches with descendants
1.0
0.8
0.6
0.4
0.2
0.0
Number of saltational branches with mutation
2
3
4
5
6
7
8
9
10
15
20
30
40
Gene
ORF1a
ORF1b
S
ORF3a
E
M
ORF7a
ORF8
N
ORF9b
1000L
477N
440K
764K
82T
153T
478K
452R
190S
681H
203K
452Q
501Y
257S
484E
204G
368I
210V
460K
3255I
859N
245N
3653F
356T
936H
3164H
490S
450D
484K
2158I
158S
64R
5V
213E
152R
2980N
3F
4L
67A
452M
1795Q
9L
501T
13P
481K
478T
493Q
102K
10P
498R
449H
486F
346R
440R
120L
4165A
49P
417T
339G
403K
681R
446G
118V
1315R
385I
478E
375S
1638N
1323G
1638I
340D
417I
954Q
4097K
83V
30I
376T
1639N
1322I